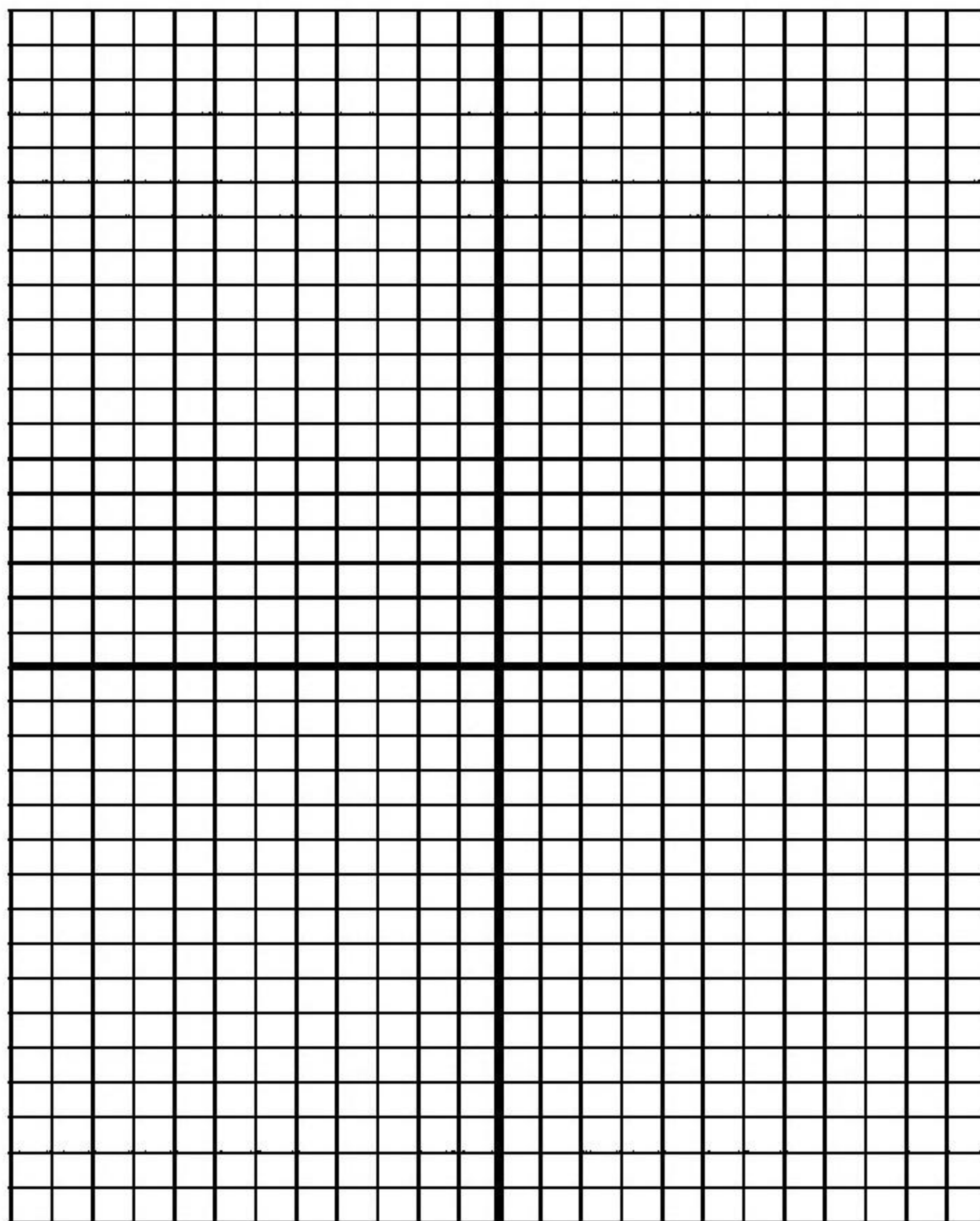
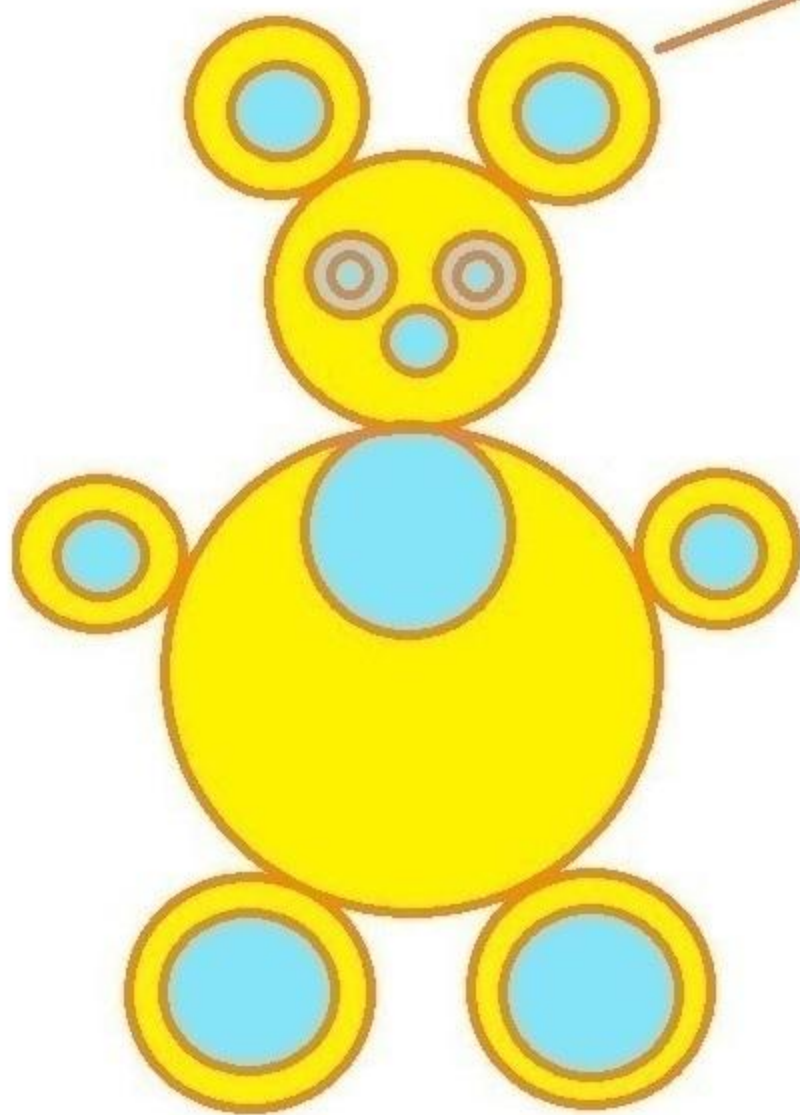


Name: _____ Period: _____ Date: _____

SKETCHING THE GRAPH OF THE EQUATION OF A CIRCLE

Materials Needed: Compass with a lead pencil or colored pencil and crayons or markers.

Hi ! I am Teddy the Bear. As you can see, I am made up of circles. Your task now is to draw me on the graph to your right. You can do that by using the EQUATION of a CIRCLE. From the given equation, just find the CENTER and the RADIUS. Then, you are all set to draw the circles. YOU CAN DO IT. Have fun !!!



In order to draw the figure, find the CENTER and RADIUS of the given EQUATION of a CIRCLE for each part of Teddy the Bear.

HEAD

$$(x-0)^2 + (y-11)^2 = 16$$

Center: _____

Radius: _____

BODY (It has an outer circle and an inner circle)

Outer Circle

$$x^2 + y^2 = 49$$

Center: _____

Radius: _____

Inner Circle

$$(x-0)^2 + (y-4)^2 = 9$$

Center: _____

Radius: _____

EARS (Each has an inner circle and an outer circle)

RIGHT EAR

Outer Circle

$$(x-3)^2 + (y-16)^2 = 4$$

Center: _____

Radius: _____

Inner Circle

$$(x-3)^2 + (y-16)^2 = 1$$

Center: _____

Radius: _____

LEFT EAR

Outer Circle

$$(x+3)^2 + (y-16)^2 = 4$$

Center: _____

Radius: _____

Inner Circle

$$(x+3)^2 + (y-16)^2 = 1$$

Center: _____

Radius: _____

EYES (Each has an inner circle and an outer circle)

RIGHT EYE

Outer Circle

$$(x-2)^2 + (y-12)^2 = 1$$

Center: _____

Radius: _____

Inner Circle

$$(x-2)^2 + (y-12)^2 = 0.25$$

Center: _____

Radius: _____

LEFT EYE

Outer Circle

$$(x+2)^2 + (y-12)^2 = 1$$

Center: _____

Radius: _____

Inner Circle

$$(x+2)^2 + (y-12)^2 = 0.25$$

Center: _____

Radius: _____

NOSE

$$x^2 + (y-10)^2 = 1$$

Center: _____

Radius: _____

HANDS (Each has an inner circle and an outer circle)

RIGHT HAND

Outer Circle

$$(x-8)^2 + (y-4)^2 = 4$$

Center: _____

Radius: _____

Inner Circle

$$(x-8)^2 + (y-4)^2 = 1$$

Center: _____

Radius: _____

LEFT HAND

Outer Circle

$$(x+8)^2 + (y-4)^2 = 4$$

Center: _____

Radius: _____

Inner Circle

$$(x+8)^2 + (y-4)^2 = 1$$

Center: _____

Radius: _____

FEET (Each has an inner circle and an outer circle)

RIGHT FOOT

Outer Circle

$$(x-4)^2 + (y+9)^2 = 9$$

Center: _____

Radius: _____

Inner Circle

$$(x-4)^2 + (y+9)^2 = 4$$

Center: _____

Radius: _____

LEFT FOOT

Outer Circle

$$(x+4)^2 + (y+9)^2 = 9$$

Center: _____

Radius: _____

Inner Circle

$$(x+4)^2 + (y+9)^2 = 4$$

Center: _____

Radius: _____

For ALGEBRA 2 Students, these are the given equations for FEET below:

Show your solution at the space provided.

FEET (Each has an inner circle and an outer circle)

Show Solution at the space below:

RIGHT FOOT

Outer Circle

$$x^2 + y^2 - 8x + 18y = -88$$

Center: _____

Radius: _____

Inner Circle

$$x^2 + y^2 - 8x + 18y = -93$$

Center: _____

Radius: _____

LEFT FOOT

Outer Circle

$$x^2 + y^2 + 8x + 18y = -88$$

Center: _____

Radius: _____

Inner Circle

$$x^2 + y^2 + 8x + 18y = -93$$

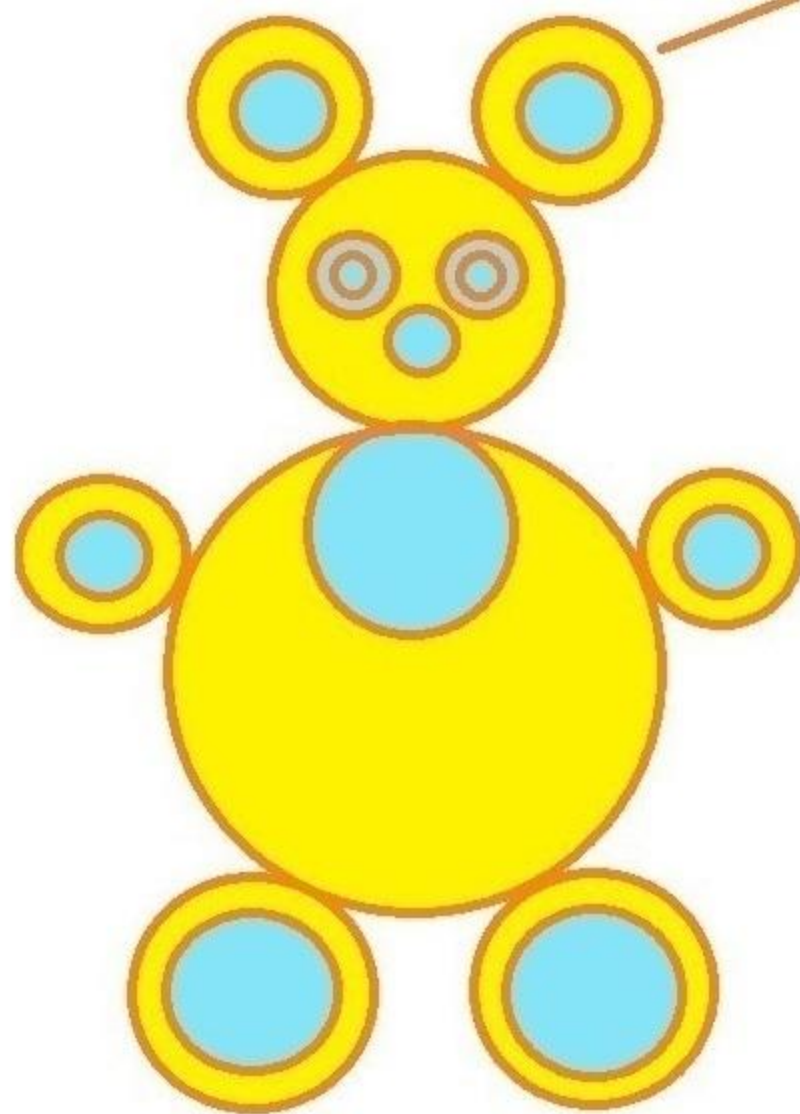
Center: _____

Radius: _____

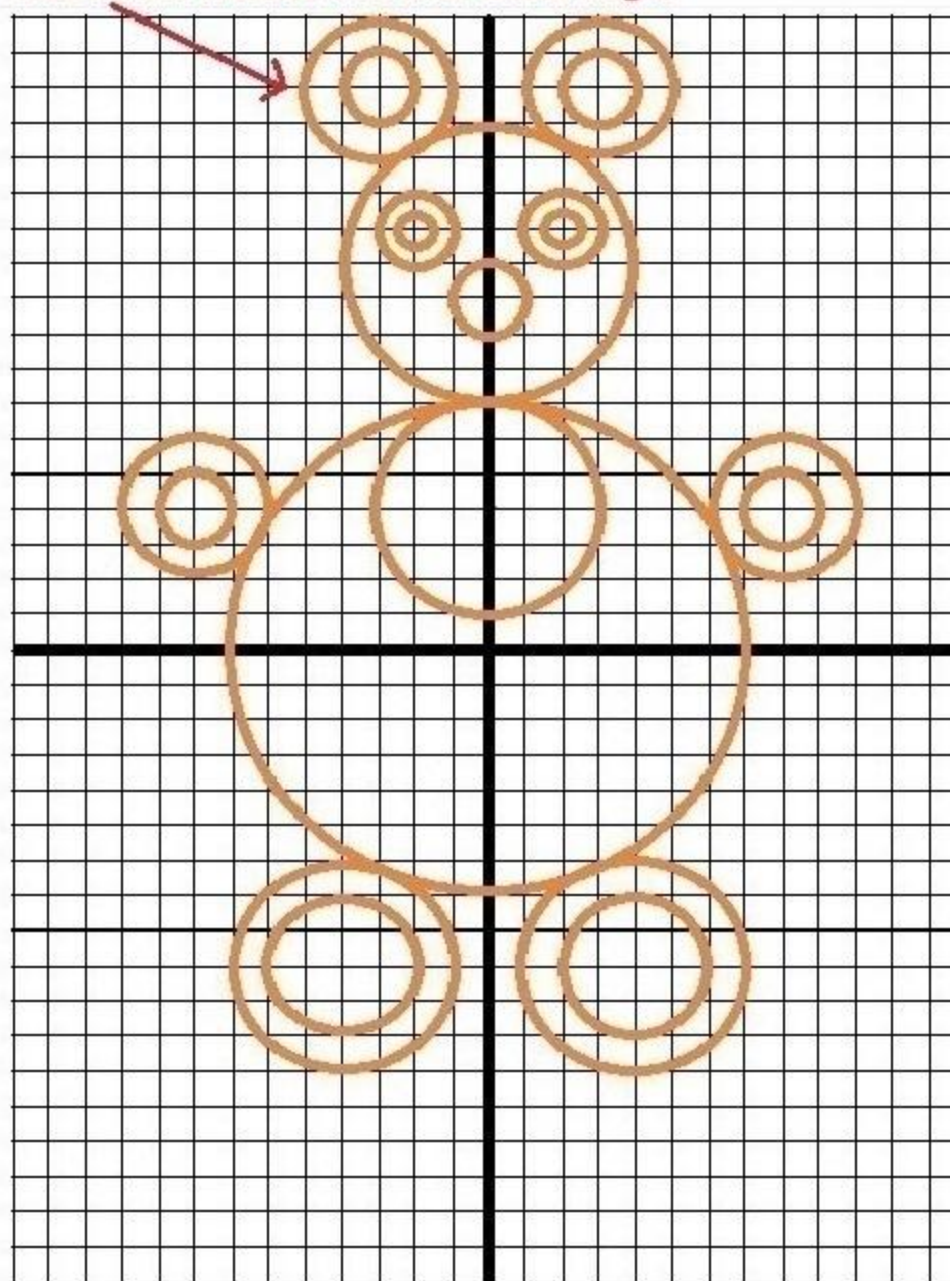
Name: ANSWER KEY Period: Date:

SKETCHING THE GRAPH OF THE EQUATION OF A CIRCLE

Hi ! I am Teddy the Bear. As you can see, I am made up of circles. Your task now is to draw me on the graph to your right. You can do that by using the EQUATION of a CIRCLE. From the given equation, just find the CENTER and the RADIUS. Then, you are all set to draw the circles. YOU CAN DO IT. Have fun !!!



This is how it looks like after sketching it



In order to draw the figure, find the CENTER and RADIUS of the given EQUATION of a CIRCLE for each part of Teddy the Bear.

HEAD

$$(x-0)^2 + (y-11)^2 = 16$$

Center: (0, 11)

Radius: 4

BODY (It has an outer circle and an inner circle)

Outer Circle

$$x^2 + y^2 = 49$$

Center: (0,0)

Radius: 7

Inner Circle

$$(x-0)^2 + (y-4)^2 = 9$$

Center: (0, 4)

Radius: 3

EARS (Each has an inner circle and an outer circle)

RIGHT EAR

Outer Circle

$$(x-3)^2 + (y-16)^2 = 4$$

Center: (3, 16)

Radius: 2

Inner Circle

$$(x-3)^2 + (y-16)^2 = 1$$

Center: (3, 16)

Radius: 1

LEFT EAR

Outer Circle

$$(x+3)^2 + (y-16)^2 = 4$$

Center: (-3, 16)

Radius: 2

Inner Circle

$$(x+3)^2 + (y-16)^2 = 1$$

Center: (-3, 16)

Radius: 1

EYES (Each has an inner circle and an outer circle)

RIGHT EYE

Outer Circle

$$(x-2)^2 + (y-12)^2 = 1$$

Center: (2, 12)

Radius: 1

Inner Circle

$$(x-2)^2 + (y-12)^2 = 0.25$$

Center: (2, 12)

Radius: 0.5

LEFT EYE

Outer Circle

$$(x+2)^2 + (y-12)^2 = 1$$

Center: (-2, 12)

Radius: 1

Inner Circle

$$(x+2)^2 + (y-12)^2 = 0.25$$

Center: (-2, 12)

Radius: 0.5

NOSE

$$x^2 + (y-10)^2 = 1$$

Center: (0, 10)

Radius: 1

HANDS (Each has an inner circle and an outer circle)

RIGHT HAND

Outer Circle

$$(x-8)^2 + (y-4)^2 = 4$$

Center: (8, 4)

Radius: 2

Inner Circle

$$(x-8)^2 + (y-4)^2 = 1$$

Center: (8, 4)

Radius: 1

LEFT HAND

Outer Circle

$$(x+8)^2 + (y-4)^2 = 4$$

Center: (-8, 4)

Radius: 2

Inner Circle

$$(x+8)^2 + (y-4)^2 = 1$$

Center: (-8, 4)

Radius: 1

FEET (Each has an inner circle and an outer circle)

RIGHT FOOT

Outer Circle

$$(x-4)^2 + (y+9)^2 = 9$$

Center: (4, -9)

Radius: 3

Inner Circle

$$(x-4)^2 + (y+9)^2 = 4$$

Center: (4, -9)

Radius: 2

LEFT FOOT

Outer Circle

$$(x+4)^2 + (y+9)^2 = 9$$

Center: (-4, -9)

Radius: 3

Inner Circle

$$(x+4)^2 + (y+9)^2 = 4$$

Center: (-4, -9)

Radius: 2

For ALGEBRA 2 Students, these are the given equations for FEET below:

Show your solution at the space provided.

FEET (Each has an inner circle and an outer circle)

Show Solution at the space below:

RIGHT FOOT

Outer Circle

$$x^2 + y^2 - 8x + 18y = -88$$

Center: (4, -9)

Radius: 3

$$x^2 - 8x + \underline{\hspace{1cm}} + y^2 + 18y + \underline{\hspace{1cm}} = -88$$

$$x^2 - 8x + (-4)^2 + y^2 + 18y + (9)^2 = -88 + (-4)^2 + (9)^2$$

$$x^2 - 8x + 16 + y^2 + 18y + 81 = -88 + 16 + 81$$

$$(x-4)^2 + (y+9)^2 = 9$$

Center: (4, -9) Radius: 3

Inner Circle

$$x^2 + y^2 - 8x + 18y = -93$$

Center: (4, -9)

Radius: 2

$$x^2 - 8x + \underline{\hspace{1cm}} + y^2 + 18y + \underline{\hspace{1cm}} = -93$$

$$x^2 - 8x + (-4)^2 + y^2 + 18y + (9)^2 = -93 + (-4)^2 + (9)^2$$

$$x^2 - 8x + 16 + y^2 + 18y + 81 = -93 + 16 + 81$$

$$(x-4)^2 + (y+9)^2 = 4$$

Center: (4, -9) Radius: 2

LEFT FOOT

Outer Circle

$$x^2 + y^2 + 8x + 18y = -88$$

Center: (-4, -9)

Radius: 3

$$x^2 + 8x + \underline{\hspace{1cm}} + y^2 + 18y + \underline{\hspace{1cm}} = -88$$

$$x^2 + 8x + (4)^2 + y^2 + 18y + (9)^2 = -88 + (4)^2 + (9)^2$$

$$x^2 + 8x + 16 + y^2 + 18y + 81 = -88 + 16 + 81$$

$$(x+4)^2 + (y+9)^2 = 9$$

Center: (-4, -9) Radius: 3

Inner Circle

$$x^2 + y^2 + 8x + 18y = -93$$

Center: (-4, -9)

Radius: 2

$$x^2 + 8x + \underline{\hspace{1cm}} + y^2 + 18y + \underline{\hspace{1cm}} = -93$$

$$x^2 + 8x + (-4)^2 + y^2 + 18y + (9)^2 = -93 + (-4)^2 + (9)^2$$

$$x^2 + 8x + 16 + y^2 + 18y + 81 = -93 + 16 + 81$$

$$(x+4)^2 + (y+9)^2 = 4$$

Center: (-4, -9) Radius: 2